

Formulas for Derivatives

1. $\frac{d}{dx} (x^n) = nx^{n-1}$
2. $\frac{d}{dx} [f(g(x))] = f'(g(x)) \cdot g'(x)$ (Chain rule)
3. $\frac{d}{dx} (f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$
4. $\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}$
5. $\frac{d}{dx} (\sin ax) = a \cos ax$
6. $\frac{d}{dx} (\cos ax) = -a \sin ax$
7. $\frac{d}{dx} e^x = e^x$
8. $\frac{d}{dx} e^{u(x)} = u'(x)e^{u(x)}$
9. $\frac{d}{dx} (b^x) = b^x \ln b$
10. $\frac{d}{dx} (b^{u(x)}) = b^{u(x)} (\ln b) u'(x)$
11. $\frac{d}{dx} \ln x = \frac{1}{x}$
12. $\frac{d}{dx} (\log_b x) = \frac{1}{x \ln b}$, for $x > 0$

Formulas for Integration

1. $\int x^n dx = \frac{1}{n+1} x^{n+1} + C$; $n \neq -1$
2. $\int \cos ax dx = \frac{1}{a} \sin ax + C$
3. $\int \sin ax dx = -\frac{1}{a} \cos ax + C$
4. $\int e^{ax} dx = \frac{1}{a} e^{ax} + C$
5. $\int \frac{1}{x} dx = \ln |x| + C$
6. $\int b^x dx = \frac{1}{\ln b} b^x + C$ for $b > 0$ and $b \neq 1$
7. $\int \ln x dx = x \ln x - x + C$
8. $\int \tan x dx = \ln |\cos x| + C$
9. $\int \sec x dx = \ln |\sec x + \tan x| + C$
10. $\int f(g(x))g'(x) dx = \int f(u) du$ (Substitution rule)
11. $\int u dv = uv - \int v du$ (Integration by parts)

General Reduction Formulas

1. $\int \sin^n ax dx = -\frac{\sin^{n-1} ax \cdot \cos ax}{an} + \frac{n-1}{n} \int \sin^{n-2} ax dx$
2. $\int \cos^n ax dx = \frac{\cos^{n-1} ax \cdot \sin ax}{an} + \frac{n-1}{n} \int \cos^{n-2} ax dx$

Trigonometric Substitution Identities

1. $a^2 - x^2 \rightarrow a^2 \cos^2 \theta$ where $x = a \sin \theta$

2. $a^2 + x^2 \rightarrow a^2 \sec^2 \theta$ where $x = a \tan \theta$

3. $x^2 - a^2 \rightarrow a^2 \tan^2 \theta$ where $x = a \sec \theta$

Pythagorean Identity

4. $\sin^2 \theta + \cos^2 \theta = 1$

Half-Angle Formulas

5. $\cos^2 \theta = \frac{1+\cos 2\theta}{2}; \quad \sin^2 \theta = \frac{1-\cos 2\theta}{2}$

Double-Angle Identities

6. $\sin 2\theta = 2 \sin \theta \cos \theta; \quad \cos 2\theta = 2 \cos^2 \theta - 1$

images/Trig_Sub_reference_Triangle_a^2-x^2.png

General Slicing Method. Suppose a solid object extends from $x = a$ to $x = b$ and the cross section of the solid perpendicular to the x -axis has an area given by a function A that is integrable on $[a, b]$. The volume of the solid is

$$V = \int_a^b A(x) dx$$

Disk Method About the x -Axis. Let f be continuous with $f(x) \geq 0$ on the interval $[a, b]$. If the region R bounded by the graph of f , the x -axis, and the lines $x = a$ and $x = b$ is revolved about the x -axis, the volume of the resulting solid of revolution is

$$V = \int_a^b \pi f(x)^2 dx.$$

Arc length. Let f have a continuous first derivative on the interval $[a, b]$. The length of the curve from $(a, f(a))$ to $(b, f(b))$ is

$$L = \int_a^b \sqrt{1 + f'(x)^2} dx.$$